

QR Factorization by Gram-Schmidt

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Introduction

- A QR decomposition of a matrix A is a decomposition of A as $A = QR$, where Q is an orthogonal matrix i.e. $Q^T Q = Q Q^T = I$ and R is an upper triangular matrix.
- There are several methods for computing the QR factorization like Householder reflections, Given Rotations, Gram-Schmidt Process. Here we will discuss the Gram-Schmidt process.

Gram-Schmidt Process

Theorem

Let $\{x_1, x_2, \dots, x_n\}$ be a basis for the inner product space V . Let $u_1 = \left(\frac{1}{\|x_1\|}\right)x_1$ and define $u_1, u_2, \dots, u_n$ recursively by $u_{k+1} = \frac{1}{\|x_{k+1} - p_k\|}(x_{k+1} - p_k)$ for $k = 1, \dots, n-1$ where $p_k = \langle x_{k+1}, u_1 \rangle u_1 + \langle x_{k+1}, u_2 \rangle u_2 + \dots + \langle x_{k+1}, u_k \rangle u_k$ is the projection of x_{k+1} onto $\text{Span}(u_1, u_2, \dots, u_k)$. Then the set $\{u_1, u_2, \dots, u_n\}$ is an orthonormal basis for V .

Gram-Schmidt QR Factorization

Theorem

If A is an $m \times n$ matrix of rank n , then A can be factored into a product QR , where Q is an $m \times n$ matrix with orthonormal column vectors and R is an upper triangular $n \times n$ matrix whose diagonal entries are all positive. Note that R must be nonsingular, since $\det(R) > 0$.

Proof

Let $\vec{p}_1, \dots, \vec{p}_n$ be the projection vectors (As defined in previous slide) and let $\{\vec{q}_1, \vec{q}_2, \dots, \vec{q}_n\}$ be the orthonormal basis of $R(A)$ (Column space of A) derived from the Gram-Schmidt process. Define

$$r_{11} = \|\vec{a}_1\|$$

$$r_{kk} = \|\vec{a}_k - \vec{p}_{k-1}\| \text{ for } k = 2, \dots, n \text{ and}$$

$$r_{ik} = \vec{q}_i^T \vec{a}_k \text{ for } i = 1, \dots, k-1 \text{ and } k = 2, \dots, n$$

Gram-Schmidt QR Factorization (Continued)

Proof(Continued)

By the Gram-Schmidt process,

$$r_{11}\vec{q}_1 = \vec{a}_1$$

$$r_{kk}\vec{q}_k = \vec{a}_k - r_{1k}\vec{q}_1 - \dots - r_{k-1,k}\vec{q}_{k-1} \text{ for } k = 2, \dots, n$$

The system obtained from Gram-Schmidt Process may be rewritten in the form

$$\begin{aligned}\vec{a}_1 &= r_{11}\vec{q}_1 \\ \vec{a}_2 &= r_{12}\vec{q}_1 + r_{22}\vec{q}_2 \\ &\vdots \\ \vec{a}_n &= r_{1n}\vec{q}_1 + \dots + r_{nn}\vec{q}_n\end{aligned}$$

Gram-Schmidt QR Factorization (Continued)

Proof(Continued)

If we set $Q = (\vec{q}_1, \vec{q}_2, \dots, \vec{q}_n)$

and define R to be the upper triangular matrix

$$\begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ 0 & r_{22} & \dots & r_{2n} \\ \vdots & & & \\ 0 & 0 & \dots & r_{nn} \end{bmatrix}$$

then the j th column of the product QR will be

$$Qr_j = r_{1j}\vec{q}_1 + r_{2j}\vec{q}_2 + \dots + r_{jj}\vec{q}_j = \vec{a}_j$$

for $j = 1, \dots, n$. Therefore,

$$QR = (\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n) = A \text{ (Proved)}$$

Application

If the columns of an $m \times n$ matrix A form an orthonormal set, then the least squares solution of $A\vec{x} = \vec{b}$ is simply $\hat{x} = A^T \vec{b}$. If A has rank n , but its column vectors do not form an orthonormal set in $\mathbb{R}^m$, then the Gram-Schmidt QR factorization can be used to solve the least squares problem. Here is an important theorem regarding this.

Theorem

If A is an $m \times n$ matrix of rank n , then the least squares solution of $A\vec{x} = \vec{b}$ is given by $\hat{x} = R^{-1}Q^T \vec{b}$, where Q and R are the matrices obtained from the factorization of A discussed before. The solution $\hat{x}$ may be obtained by using back substitution to solve $R\vec{x} = Q^T \vec{b}$.

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